

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12411334
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Chen; Ping-Yi et al.

Optical lens assembly and head-mounted electronic device

Abstract

An optical lens assembly includes, an optical element group; and in order from a visual side to an image source side: a first lens with positive refractive power, a second lens with refractive power, a third lens with refractive power, and an image source-side surface of the third lens being convex in a paraxial region thereof, and a partial-reflective-partial-transmissive element. Wherein the optical element group is disposed between the first lens and the third lens, and includes, in order from the visual side to the image source side: an absorptive polarizer, a reflective polarizer and a phase retarder. The optical lens assembly may become lightweight and have good image quality when satisfying a specific condition.

Inventors: Chen; Ping-Yi (Taichung, TW), Tsai; Fei-Hsin (Taichung, TW), Ge; Cong (Guangdong Province, CN)

Applicant: NEWMAX TECHNOLOGY CO., LTD. (Taichung, TW)

Family ID: 89722873

Appl. No.: 18/220279

Filed: July 11, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240345380 A1	Oct. 17, 2024

Foreign Application Priority Data

TW	112113797	Apr. 13, 2023
----	-----------	---------------

Publication Classification

Int. Cl.: G02B25/00 (20060101); G02B27/01 (20060101)

U.S. Cl.:

CPC **G02B25/001** (20130101); **G02B27/0172** (20130101);

Field of Classification Search

CPC: G02B (25/001); G02B (27/0172)

USPC: 359/630; 359/716; 359/784

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2023/0400694	12/2022	Yamada	N/A	G02F 1/1335

Primary Examiner: Choi; William

Attorney, Agent or Firm: Bruce Stone LLP

Background/Summary

BACKGROUND

Field of the Invention

(1) The present invention relates to an optical lens assembly and head-mounted electronic device, and more particularly to an optical lens assembly applicable to head-mounted electronic devices.

Description of Related Art

(2) With the development of the semiconductor industry, the functions of various consumer electronic products are increasingly powerful. Moreover, various services of the software application end emerge. These enable consumers to have more choices. Virtual reality (VR) technology, Augmented reality (AR) technology and Mixed reality (MR) technology emerge when the market is no longer satisfied with handheld electronic products. Nowadays, the application of virtual reality opens up a blue ocean market for consumer electronics, and in the application field of virtual reality, the first commercialized project is the head-mounted display.

(3) However, the current head-mounted displays are heavy and have poor image quality.

(4) The present invention mitigates and/or obviates the aforementioned disadvantages.

SUMMARY

(5) The objective of the present invention is to provide an optical lens assembly and a head-mounted electronic device, which can reduce the number of lenses by folding the light path, so as to reduce the weight of the device, and also can provide better image quality.

(6) Therefore, an optical lens assembly in accordance with an embodiment of the present invention includes: an optical element group; and in order from a visual side to an image source side: a first lens with positive refractive power, a second lens with refractive power, a third lens with refractive power, and an image source-side surface of the third lens being convex in a paraxial region thereof, and a partial-reflective-partial-transmissive element. Wherein the optical element group is disposed between the first lens and the third lens, and includes, in order from the visual side to the image source side: an absorptive polarizer (that is, a first absorptive polarizer), a reflective polarizer and a phase retarder (that is, a first phase retarder). A radius of curvature of an image source-side surface

of the first lens is $R2$, a thickness of the first lens along an optical axis is $CT1$, a focal length of the first lens is $f1$, a focal length of the optical lens assembly is f , and the following condition is satisfied: $-5.88 < (R2/CT1) * (f1/f) < 110.22$.

(7) When $-5.88 < (R2/CT1) * (f1/f) < 110.22$ is satisfied, the distortion of the optical lens assembly can be effectively improved, the aberration of the optical lens assembly can be reduced, and the size of the lens can be further reduced.

(8) Optionally, a focal length of the third lens is $f3$, the focal length of the optical lens assembly is f , and the following condition is satisfied: $-6.92 < f3/f < 23.80$, so that the ratio of the focal length of the third lens to that of the optical lens assembly can enhance the wide-field of view characteristic of the optical lens assembly, thereby providing a larger field of view and maintaining the illumination of the optical lens assembly.

(9) Optionally, the focal length of the third lens is $f3$, a radius of curvature of a visual-side surface of the first lens is $R1$, and the following condition is satisfied: $-5.58 < f3/R1 < 27.18$, which can effectively improve the aberration of the optical lens assembly, further improving the image quality.

(10) Optionally, a maximum effective radius of the visual-side surface of the first lens is $CA1$, an absolute value of a displacement in parallel to the optical axis from an intersection between the visual-side surface of the first lens and the optical axis to the maximum effective radius position on the visual-side surface of the first lens is $TDP1$, and the following condition is satisfied:

$1.76 < CA1/TDP1 < 137.17$, which can achieve a larger field of view and reduce the size of the lens.

(11) Optionally, a maximum effective radius of the image source-side surface of the first lens is $CA2$, an absolute value of a displacement in parallel to the optical axis from an intersection between the image source-side surface of the first lens and the optical axis to the maximum effective radius position on the image source-side surface of the first lens is $TDP2$, and the following condition is satisfied: $1.43 < CA2/TDP2 < 56.16$, which can achieve a larger field of view and reduce the size of the lens.

(12) Optionally, a radius of curvature of the image source-side surface of the third lens is $R6$, a thickness of the third lens along the optical axis is $CT3$, and the following condition is satisfied: $-58.44 < R6/CT3 < -5.15$, which is conducive to achieving a proper balance between the radius of curvature and the thickness of the third lens.

(13) Optionally, the radius of curvature of the visual-side surface of the first lens is $R1$, the radius of curvature of the image source-side surface of the first lens is $R2$, the focal length of the first lens is $f1$, and the following condition is satisfied: $-32.71 \text{ mm} < (R1 * R2)/f1 < 60.84 \text{ mm}$, which is conducive to achieving the more appropriate distribution of the radius of curvature of the first lens, thereby reducing the chromatic aberration.

(14) Optionally, the radius of curvature of the visual-side surface of the first lens is $R1$, the radius of curvature of the image source-side surface of the first lens is $R2$, and the following condition is satisfied: $-3.55 < R1/R2 < 6.52$, which is conducive to preventing the radius of curvature from being too small and reducing the sensitivity to the assembly tolerance as the two radii of curvature of the first lens are conditioned by each other.

(15) Optionally, the radius of curvature of the image source-side surface of the third lens is $R6$, the radius of curvature of the image source-side surface of the first lens is $R2$, and the following condition is satisfied: $-4.59 < R6/R2 < 6.08$, which is conducive to preventing the radius of curvature from being too small and reducing the sensitivity to the assembly tolerance as the two radii of curvature are conditioned by each other.

(16) Optionally, a thickness of the second lens along the optical axis is $CT2$, the thickness of the third lens along the optical axis is $CT3$, and the following condition is satisfied:

$0.22 < CT3/CT2 < 7.58$, which is conducive to ensuring that the thickness of the optical lens assembly can meet the processing requirement of the manufacturing process of the lens device, while ensuring the image quality.

(17) Optionally, the thickness of the first lens along the optical axis is $CT1$, the thickness of the

second lens along the optical axis is CT2, the thickness of the third lens along the optical axis is CT3, and the following condition is satisfied: $0.62 < (CT1 + CT2) / CT3 < 8.67$, which is conducive to improving the thickness matching between the lenses and ensuring the assembly stability, while ensuring the image quality.

(18) Optionally, the radius of curvature of the visual-side surface of the first lens is R1, the thickness of the first lens along the optical axis is CT1, and the following condition is satisfied: $-13.48 < R1 / CT1 < 9.76$, so that the distortion of the optical lens assembly can be effectively improved, the aberration of the optical lens assembly can be reduced, and the size of the lens can be further reduced.

(19) Optionally, the focal length of the first lens is f1, the focal length of the optical lens assembly is f, and the following condition is satisfied: $1.42 < f1 / f < 16.1$, so that the ratio of the focal length of the first lens to that of the optical lens assembly can enhance the wide-field of view characteristic of the optical lens assembly, thereby providing a larger field of view and maintaining the illumination of the optical lens assembly.

(20) Optionally, the maximum effective radius of the image source-side surface of the first lens is CA2, a distance from the visual-side surface of the first lens to an image source plane along the optical axis is TL, the radius of curvature of the visual-side surface of the first lens is R1, and the following condition is satisfied: $-49.53 \text{ mm} < (CA2 / TL) * R1 < 43.8 \text{ mm}$, which is conducive to achieving a proper balance between the radius of curvature, the size of the first lens, and the length of the optical lens assembly.

(21) Optionally, the focal length of the first lens is f1, the focal length of the third lens is f3, and the following condition is satisfied: $-1.10 < f1 / f3 < 3.53$, which is conducive to achieving the more appropriate distribution of the refractive power of the optical lens assembly, thereby reducing the aberration.

(22) Optionally, a focal length of the second lens is f2, the focal length of the optical lens assembly is f, and the following condition is satisfied: $-5.80 < f2 / f < 13.35$, so that the ratio of the focal length of the second lens to that of the optical lens assembly can enhance the wide-field of view characteristic of the optical lens assembly, thereby providing a larger field of view and maintaining the illumination of the optical lens assembly.

(23) Optionally, the radius of curvature of the image source-side surface of the third lens is R6, a radius of curvature of a visual-side surface of the third lens is R5, and the following condition is satisfied: $-1.37 < R6 / R5 < 3.67$, which is conducive to preventing the radius of curvature from being too small and reducing the sensitivity to the assembly tolerance as the two radii of curvature of the third lens are conditioned by each other.

(24) Moreover, a head-mounted electronic device in accordance with an embodiment of the present invention includes a housing, the aforementioned optical lens assembly disposed in the housing, an image source disposed on the image source plane of the optical lens assembly in the housing, and a controller disposed in the housing and electrically connected to the image source.

(25) The present invention will be presented in further details from the following descriptions with the accompanying drawings, which show, for purpose of illustrations only, the preferred embodiments in accordance with the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a schematic view of an optical lens assembly in accordance with a first embodiment of the present invention;
- (2) FIG. 1B is a schematic diagram of a part of the optical lens assembly of FIG. 1A with a light path of a chief ray therein;

- (3) FIG. 1 is a partial amplified view of FIG. 1A;
- (4) FIG. 2 is a schematic view of an optical lens assembly in accordance with a second embodiment of the present invention;
- (5) FIG. 3 is a schematic view of an optical lens assembly in accordance with a third embodiment of the present invention;
- (6) FIG. 4 is a schematic view of an optical lens assembly in accordance with a fourth embodiment of the present invention;
- (7) FIG. 5 is a schematic view of an optical lens assembly in accordance with a fifth embodiment of the present invention;
- (8) FIG. 6 is a schematic view of an optical lens assembly in accordance with a sixth embodiment of the present invention;
- (9) FIG. 7 is a schematic view of an optical lens assembly in accordance with a seventh embodiment of the present invention;
- (10) FIG. 8 is a schematic view of an optical lens assembly in accordance with an eighth embodiment of the present invention; and
- (11) FIG. 9 is a schematic diagram of a head-mounted electronic device in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION

First Embodiment

- (12) Referring to FIGS. 1A, 1B and 1C, an optical lens assembly in accordance with a first embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis **190**: a stop **100**, a first lens **110**, a first optical element group **140**, a second lens **120**, a third lens **130**, a partial-reflective-partial-transmissive element **150**, a second optical element group **160** and an image source plane **181**. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.
- (13) The stop **100** may be located in a position where the user's eyes view an image.
- (14) The first lens **110** with positive refractive power includes a visual-side surface **111** and an image source-side surface **112**, the visual-side surface **111** of the first lens **110** is convex in a paraxial region thereof, the image source-side surface **112** of the first lens **110** is concave in a paraxial region thereof, the visual-side surface **111** and the image source-side surface **112** of the first lens **110** are aspheric, and the first lens **110** is made of plastic.
- (15) The second lens **120** with positive refractive power includes a visual-side surface **121** and an image source-side surface **122**, the visual-side surface **121** of the second lens **120** is flat in a paraxial region thereof, the image source-side surface **122** of the second lens **120** is convex in a paraxial region thereof, the image source-side surface **122** of the second lens **120** is spherical, and the second lens **120** is made of plastic.
- (16) The third lens **130** with positive refractive power includes a visual-side surface **131** and an image source-side surface **132**, the visual-side surface **131** of the third lens **130** is concave in a paraxial region thereof, the image source-side surface **132** of the third lens **130** is convex in a paraxial region thereof, the visual-side surface **131** and the image source-side surface **132** of the third lens **130** are aspheric, and the third lens **130** is made of plastic.
- (17) The first optical element group **140** is disposed on the visual-side surface **121** of the second lens **120** and includes, in order from the visual side to the image source side along the optical axis **190**: a first absorptive polarizer **141**, a reflective polarizer **142** and a first phase retarder **143**. The first phase retarder **143** is disposed on the visual-side surface **121** of the second lens **120** and is, for example, but not limited to, a quarter-wave plate. The reflective polarizer **142** is disposed on the first phase retarder **143**. The first absorptive polarizer **141** is disposed on the reflective polarizer **142**. The first optical element group **140** is located between the first lens **110** and the third lens **130**.
- (18) The partial-reflective-partial-transmissive element **150** is disposed on the image source-side surface **132** of the third lens **130** and has an average reflectance of at least 30%, preferably 50%, in

the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **150** for different wavelengths.

(19) The second optical element group **160** is disposed on the image source plane **181** and includes, in order from the visual side to the image source side along the optical axis **190**: a second phase retarder **161** and a second absorptive polarizer **162**. The second absorptive polarizer **162** is disposed on the image source plane **181**. The second phase retarder **161** is disposed on the second absorptive polarizer **162** and is, for example, but not limited to, a quarter-wave plate.

(20) The optical lens assembly works in cooperation with an image source **183** disposed on the image source plane **181**. In the present embodiment, the type of the image source **183** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

(21) The curve equation for the aspheric surface profiles of the respective lenses of the first embodiment is expressed as follows:

$$(22) z(h) = \frac{ch^2}{1 + [1 - (k + 1)c^2 h^2]^{0.5}} + \text{.Math.}(A_i) \cdot \text{.Math.}(h^i)$$

wherein: z represents the value of a reference position at a height of h with respect to a vertex of the surface of a lens along the optical axis **190**; c represents a paraxial curvature (i.e., a curvature of a lens surface in a paraxial region thereof) equal to 1/R (R: a paraxial radius of curvature); h represents a vertical distance from the point on the curve of the aspheric surface to the optical axis **190**; k represents the conic constant; and Ai represents the i-th order aspheric coefficient.

(23) The optical lens assembly of the first embodiment utilizes the configuration and arrangement of the absorptive polarizer, the reflective polarizer, the phase retarders, the partial-reflective-partial-transmissive element and the lenses to fold the light path thereof by the transmission and reflection of light to shorten the length of the optical lens assembly required for forming an image without affecting the image quality. In a light path L in FIG. 1B, an unpolarized beam from the image source **183** turns to a linearly-polarized beam after passing through the second absorptive polarizer **162**, the linearly-polarized beam turns to a circularly-polarized beam after passing through the second phase retarder **161**, and the circularly-polarized beam then transmits to the partial-reflective-partial-transmissive element **150**. A component of the circularly-polarized beam passes through the partial-reflective-partial-transmissive element **150**, and then passes through the third lens **130**, the second lens **120** and the first phase retarder **143** sequentially to turn to a linearly-polarized beam with a polarization direction parallel to the reflective axis of the reflective polarizer **142** and transmit to the reflective polarizer **142**. This linearly-polarized beam is reflected by the reflective polarizer **142** and passes through the first phase retarder **143** again to turn to a circularly-polarized beam, and then passes through the second lens **120** and the third lens **130** sequentially to transmit to partial-reflective-partial-transmissive element **150**. Then, a component of the circularly-polarized beam is reflected by the partial-reflective-partial-transmissive element **150**, and then passes through the third lens **130**, the second lens **120** and the first phase retarder **143** sequentially to turn to a linearly-polarized beam with a polarization direction vertical to the reflective axis of the reflective polarizer **142**. Finally, the linearly-polarized beam transmits to the user's eyes after passing through the reflective polarizer **142**, the first absorptive polarizer **141** and the first lens **110** sequentially.

(24) Please refer to Tables 1-4, Table 1 shows the detailed optical data of the elements of the optical lens assembly of the first embodiment, Table 2 shows the aspheric coefficients of the aspherical surfaces of the elements of the optical lens assembly of the first embodiment, Table 3 shows the remaining parameters of the optical lens assembly of the first embodiment and the values thereof, and the values of the parameters in Tables 1 and 3 meet the conditional formulas of Table 4. A focal length of the first lens **110** is f1, a focal length of the second lens **120** is f2, a focal length of the third lens **130** is f3, a thickness of the first lens **110** along the optical axis **190** is CT1, a thickness of the second lens **120** along the optical axis **190** is CT2, a thickness of the third lens **130** along the

optical axis **190** is CT3, a maximum effective radius of the visual-side surface **111** of the first lens **110** is CA1, a maximum effective radius of the image source-side surface **112** of the first lens **110** is CA2, an absolute value of a displacement in parallel to the optical axis **190** from an intersection between the visual-side surface **111** of the first lens **110** and the optical axis to the maximum effective radius position on the visual-side surface **111** of the first lens **110** is TDP1, an absolute value of a displacement in parallel to the optical axis **190** from an intersection between the image source-side surface **112** of the first lens **110** and the optical axis **190** to the maximum effective radius position on the image source-side surface **112** of the first lens **110** is TDP2, a radius of curvature of the visual-side surface **111** of the first lens **110** is R1, a radius of curvature of the image-side surface **112** of the first lens **110** is R2, a radius of curvature of the visual-side surface of **131** the third lens **130** is R5, a radius of curvature of the image source-side surface **132** of the third lens **130** is R6, a distance from the visual-side surface **111** of the first lens **110** to the image source plane **181** along the optical axis **190** is TL.

(25) TABLE-US-00001 TABLE 1 Embodiment 1 f = 19.65 mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 52.08° Radius of Refractive Abbe Refraction/ Surface curvature Thickness/gap index (nd) number (vd) reflection 0 Stop Infinity 15.000 — — — 1 First lens 17.211 4.879 1.544 55.9 Refraction 2 27.594 3.660 — — Refraction 3 First absorptive polarizer Infinity 0.100 1.533 56.0 Refraction 4 Reflective polarizer Infinity 0.100 1.533 56.0 Refraction 5 First phase retarder Infinity 0.100 1.533 56.0 Refraction 6 Second lens Infinity 2.475 1.544 55.9 Refraction 7 -56.364 0.165 — — Refraction 8 Third lens -146.533 4.238 1.544 55.900 Refraction 9 Partial-reflective-partial- -87.756 -4.238 — — Reflection transmissive element 10 -56.364 -0.165 — — Refraction 11 Second lens -146.533 -2.475 1.544 55.900 Refraction 12 First phase retarder Infinity -0.100 1.533 56.0 Refraction 13 Reflective polarizer Infinity -0.100 1.533 56.0 Refraction 14 Reflective polarizer Infinity 0.100 — — Reflection 15 First phase retarder Infinity 0.100 1.533 56.0 Refraction 16 Second lens Infinity 2.475 1.544 55.900 Refraction 17 -56.364 0.165 — — Refraction 18 Third lens -146.533 4.238 1.544 55.9 Refraction 19 Partial-reflective-partial- -87.756 6.152 — — Refraction transmissive element 20 Second phase retarder Infinity 0.100 1.533 56.0 Refraction 21 Second absorptive Infinity 0.100 1.533 56.0 Refraction polarizer 22 Image source plane Infinity — — — — The reference wavelength is 550 nm.

(26) TABLE-US-00002 TABLE 2 Embodiment 1 Aspheric Coefficients Surface 1 2 6, 12, 16 7, 10, 17 8, 11, 18 9, 19 K: -1.2068E-02 2.3959E-01 0.0000E+00 0.0000E+00 9.0000E+01 0.0000E+00 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: -2.2912E-06 7.2200E-06 0.0000E+00 0.0000E+00 -3.7084E-05 -3.7367E-06 A6: 4.2764E-08 4.5178E-08 0.0000E+00 0.0000E+00 1.3923E-08 -1.2232E-09 A8: -7.8998E-11 -1.0456E-10 0.0000E+00 0.0000E+00 4.0522E-10 2.1979E-10 A10: -7.8962E-13 -4.1630E-13 0.0000E+00 0.0000E+00 4.2746E-12 5.3435E-13 A12: -9.0492E-16 -4.2221E-15 0.0000E+00 0.0000E+00 1.8432E-14 -2.2481E-15 A14: 8.7059E-18 -4.0336E-17 0.0000E+00 0.0000E+00 -2.2142E-17 2.5722E-17 A16: 5.5507E-21 -8.6824E-20 0.0000E+00 0.0000E+00 -1.1992E-18 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00

(27) TABLE-US-00003 TABLE 3 Embodiment 1 f1 [mm] 71.67 CA1 [mm] 13.01 TDP1 [mm] 5.91 f2 [mm] 102.99 CA2 [mm] 12.64 TDP2 [mm] 3.16 f3 [mm] 389.82 — — — —

(28) TABLE-US-00004 TABLE 4 Embodiment 1 f1/f 3.65 f3/f 19.83 R1/R2 0.62 R6/R5 0.60 R6/R2 -3.18 R6/CT3 -20.71 CT3/CT2 1.71 (R1*R2)/f1 1.22 (CT1 + CT2)/CT3 1.74 CA1/TDP1 2.20 R1/CT1 3.53 CA2/TDP2 4.01 f3/R1 22.65 (CA2/TL)*R1 9.86 f1/f3 0.18 (R2/CT1)(f1/f) 20.62 f2/f 5.24 — —

(29) In Table 1, the units of the radius of curvature, the thickness, the gap and the focal length are expressed in mm, and the surface numbers 22-0 respectively represent the surfaces to which the light sequentially transmits from the image source plane **181** to the stop **100** along the light path L,

wherein the surface 0 represents a gap between the stop **100** (or the user's eyes) and the visual-side surface **111** of the first lens **110** along the optical axis **190**; the surface 1 represents the thickness of the first lens **110** along the optical axis **190**; the surface 2 represents a gap between the image source-side surface **112** of the first lens **110** and the first absorptive polarizer **141** along the optical axis **190**; the surface 3 represents the thickness of the first absorptive polarizer **141** along the optical axis **190**; the surfaces 4, 13 and 14 represent the thickness of the reflective polarizer **142** along the optical axis **190**; the surfaces 5, 12 and 15 represent the thickness of the first phase retarder **143** along the optical axis **190**; the surfaces 6 and 16 represent the thickness of the second lens **120** along the optical axis **190**; the surface 11 represents a gap between the image source-side surface **122** of the second lens **120** and the visual-side surface **121** of the second lens **120** along the optical axis **190**, this gap corresponds to the thickness of the second lens **120** along the optical axis **190**; the surfaces 7, 10 and 17 represent a gap between the image source-side surface **122** of the second lens **120** and the visual-side surface **131** of the third lens **130** along the optical axis **190**; the surfaces 8 and 18 represent the thickness of the third lens **130** along the optical axis **190**; the surface 9 represents a gap between the image source-side surface **132** of the third lens **130** and the visual-side surface **131** of the third lens **130** along the optical axis **190**, this gap corresponds to the thickness of the third lens **130** along the optical axis **190**; the surfaces 19 represents a gap between the image source-side surface **132** of the third lens **130** and the second phase retarder **161** along the optical axis **190**; the surface 20 represents the thickness of the second phase retarder **161** along the optical axis **190**; the surface 21 represents the thickness of the second absorptive polarizer **162** along the optical axis **190**; and the surface 22 is the image source plane **181**. The gaps and thicknesses having a positive sign in Table 1 denote the transmission direction of light is toward the stop **100**, and the gaps and thicknesses having a negative sign in Table 1 denote the transmission direction of light is toward the image source plane **181**.

(30) In table 2, k represents the conic constant of the equation of aspheric surface profiles, and A_2 , A_4 , A_6 , A_8 , A_{10} , A_{12} , A_{14} , A_{16} , A_{18} , and A_{20} represent the high-order aspheric coefficients.

(31) The respective tables presented below for respective one of other embodiments are based on the schematic view of this embodiment, and the definitions of parameters in the tables are the same as those in Tables 1-4 of the first embodiment. However, the definition of each surface number in Table 1 varies with the number of the lenses and the position of the optical elements, and the relevant description of the embodiments may be referred to the definition mode of each surface number in Table 1. Therefore, an explanation in this regard will not be provided again.

Second Embodiment

(32) Referring to FIG. 2, an optical lens assembly in accordance with a second embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis **290**: a stop **200**, a first lens **210**, a first optical element group **240**, a second lens **220**, a third lens **230**, a partial-reflective-partial-transmissive element **250**, a second optical element group **260** and an image source plane **281**. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

(33) The stop **200** may be located in a position where the user's eyes view an image.

(34) The first lens **210** with positive refractive power includes a visual-side surface **211** and an image source-side surface **212**, the visual-side surface **211** of the first lens **210** is convex in a paraxial region thereof, the image source-side surface **212** of the first lens **210** is concave in a paraxial region thereof, the visual-side surface **211** and the image source-side surface **212** of the first lens **210** are aspheric, and the first lens **210** is made of plastic.

(35) The second lens **220** with negative refractive power includes a visual-side surface **221** and an image source-side surface **222**, the visual-side surface **221** of the second lens **220** is flat in a paraxial region thereof, the image source-side surface **222** of the second lens **220** is concave in a paraxial region thereof, the image source-side surface **222** of the second lens **220** is spherical, and the second lens **220** is made of plastic.

(36) The third lens **230** with positive refractive power includes a visual-side surface **231** and an image source-side surface **232**, the visual-side surface **231** of the third lens **230** is convex in a paraxial region thereof, the image source-side surface **232** of the third lens **230** is convex in a paraxial region thereof, the visual-side surface **231** of the third lens **230** is spherical, the image source-side surface **232** of the third lens **230** is aspheric, and the third lens **230** is made of plastic. The third lens **230** and the second lens **220** together form a cemented doublet lens.

(37) The first optical element group **240** is disposed on the visual-side surface **221** of the second lens **220** and includes, in order from the visual side to the image source side along the optical axis **290**: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer **141**, the reflective polarizer **142** and the first phase retarder **143** of the first embodiment and will not be explained again.

(38) The partial-reflective-partial-transmissive element **250** is disposed on the image source-side surface **232** of the third lens **230** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **250** for different wavelengths.

(39) The second optical element group **260** is disposed on the image source plane **281** and includes, in order from the visual side to the image source side along the optical axis **290**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

(40) The optical lens assembly works in cooperation with an image source **283** disposed on the image source plane **281**. In the present embodiment, the type of the image source **283** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

(41) Please refer to Tables 5-8, Table 5 shows the detailed optical data of the elements of the optical lens assembly of the second embodiment, Table 6 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the second embodiment, Table 7 shows the remaining parameters of the optical lens assembly of the second embodiment and the values thereof, and the values of the parameters in Tables 5 and 7 meet the conditional formulas of Table 8. In the second embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 5 can be referred to the relevant description of Table 1 and will not be explained again.

(42) TABLE-US-00005 TABLE 5 Embodiment 2 $f = 19.82$ mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 52.04° Radius of Thickness/ Refractive Abbe Refraction/ Surface curvature gap index (nd) number (vd) reflection 0 Stop Infinity 15.000 — — — 1 First lens 24.606 6.704 1.544 55.9 Refraction 2 126.414 4.096 — — Refraction 3 First absorptive polarizer Infinity 0.100 1.533 56.0 Refraction 4 Reflective polarizer Infinity 0.100 1.533 56.0 Refraction 5 First phase retarder Infinity 0.100 1.533 56.0 Refraction 6 Second lens Infinity 2.000 1.645 23.400 Refraction 7 Third lens 62.105 4.662 1.544 55.9 Refraction 8 Partial-reflective-partial- -71.153 -4.662 — — Reflection transmissive element 9 Second lens 62.105 -2.000 1.645 23.400 Refraction 10 First phase retarder Infinity -0.100 1.533 56.0 Refraction 11 Reflective polarizer Infinity -0.100 1.533 56.0 Refraction 12 Reflective polarizer Infinity 0.100 — — Reflection 13 First phase retarder Infinity 0.100 1.533 56.0 Refraction 14 Second lens Infinity 2.000 1.645 23.400 Refraction 15 Third lens 62.105 4.662 1.544 55.9 Refraction 16 Partial-reflective-partial- -71.153 6.038 — — Refraction transmissive element 17 Second phase retarder Infinity 0.100 1.533 56.0 Refraction 18 Second absorptive Infinity 0.100 1.533 56.0 Refraction polarizer 19 Image source plane Infinity — — — — The reference wavelength is 550 nm.

(43) TABLE-US-00006 TABLE 6 Embodiment 2 Aspheric Coefficients Surface 1 2 6, 10, 14 7, 9,

15 7, 9, 15 8, 16 K: -9.7435E+00 9.0000E+01 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00
A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: 6.4634E-05
-3.0925E-05 0.0000E+00 0.0000E+00 0.0000E+00 5.0297E-06 A6: -6.9913E-07
-3.6771E-07 0.0000E+00 0.0000E+00 0.0000E+00 1.2349E-08 A8: 4.9207E-09 1.9661E-09
0.0000E+00 0.0000E+00 0.0000E+00 1.5945E-10 A10: -1.8438E-11 -9.7058E-12
0.0000E+00 0.0000E+00 0.0000E+00 -1.0803E-12 A12: -1.5556E-13 -6.4398E-14
0.0000E+00 0.0000E+00 0.0000E+00 2.0297E-15 A14: 5.1182E-16 6.8206E-17 0.0000E+00
0.0000E+00 0.0000E+00 2.9897E-17 A16: 1.8205E-18 2.2308E-18 0.0000E+00 0.0000E+00
0.0000E+00 -1.3317E-19 A18: 5.3367E-20 5.4754E-21 0.0000E+00 0.0000E+00 0.0000E+00
0.0000E+00 A20: -4.1168E-22 -7.8393E-23 0.0000E+00 0.0000E+00 0.0000E+00
0.0000E+00

(44) TABLE-US-00007 TABLE 7 Embodiment 2 f1 [mm] 54.56 CA1 [mm] 11.07 TDP1 [mm]
1.95 f2 [mm] -95.74 CA2 [mm] 12.07 TDP2 [mm] 0.97 f3 [mm] 61.35 — — — —

(45) TABLE-US-00008 TABLE 8 Embodiment 2 f1/f 2.75 f3/f 3.10 R1/R2 0.19 R6/R5 -1.15
R6/R2 -0.56 R6/CT3 -15.26 CT3/CT2 2.33 (R1*R2)/f1 50.70 (CT1 + CT2)/CT3 1.87 CA1/TDP1
5.68 R1/CT1 3.67 CA2/TDP2 12.45 f3/R1 2.49 (CA2/TL)*R1 12.37 f1/f3 0.89 (R2/CT1)(f1/f)
51.91 f2/f -4.83 — —

Third Embodiment

(46) Referring to FIG. 3, an optical lens assembly in accordance with a third embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis **390**: a stop **300**, a first lens **310**, a first optical element group **340**, a second lens **320**, a third lens **330**, a partial-reflective-partial-transmissive element **350**, a second optical element group **360** and an image source plane **381**. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

(47) The stop **300** may be located in a position where the user's eyes view an image.

(48) The first lens **310** with positive refractive power includes a visual-side surface **311** and an image source-side surface **312**, the visual-side surface **311** of the first lens **310** is convex in a paraxial region thereof, the image source-side surface **312** of the first lens **310** is concave in a paraxial region thereof, the visual-side surface **311** and the image source-side surface **312** of the first lens **310** are aspheric, and the first lens **310** is made of plastic.

(49) The second lens **320** with positive refractive power includes a visual-side surface **321** and an image source-side surface **322**, the visual-side surface **321** of the second lens **320** is flat in a paraxial region thereof, the image source-side surface **322** of the second lens **320** is convex in a paraxial region thereof, the image source-side surface **322** of the second lens **320** is spherical, and the second lens **320** is made of plastic.

(50) The third lens **330** with negative refractive power includes a visual-side surface **331** and an image source-side surface **332**, the visual-side surface **331** of the third lens **330** is concave in a paraxial region thereof, the image source-side surface **332** of the third lens **330** is convex in a paraxial region thereof, the visual-side surface **331** of the third lens **330** is spherical, the image source-side surface **332** of the third lens **330** is aspheric, and the third lens **330** is made of plastic. The third lens **330** and the second lens **320** together form a cemented doublet lens.

(51) The first optical element group **340** is disposed on the visual-side surface **321** of the second lens **320** and includes, in order from the visual side to the image source side along the optical axis **390**: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer **141**, the reflective polarizer **142** and the first phase retarder **143** of the first embodiment and will not be explained again.

(52) The partial-reflective-partial-transmissive element **350** is disposed on the image source-side surface **332** of the third lens **330** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different

reflectance of the partial-reflective-partial-transmissive element **350** for different wavelengths.

(53) The second optical element group **360** is disposed on the image source plane **381** and includes, in order from the visual side to the image source side along the optical axis **390**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

(54) The optical lens assembly works in cooperation with an image source **383** disposed on the image source plane **381**. In the present embodiment, the type of the image source **383** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

(55) Please refer to Tables 9-12, Table 9 shows the detailed optical data of the elements of the optical lens assembly of the third embodiment, Table 10 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the third embodiment, Table 11 shows the remaining parameters of the optical lens assembly of the third embodiment and the values thereof, and the values of the parameters in Tables 9 and 11 meet the conditional formulas of Table 12. In the third embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 9 can be referred to the relevant description of Table 1 and will not be explained again.

(56) TABLE-US-00009 TABLE 9 Embodiment 3 $f = 19.78$ mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 52.05° Radius of Thickness/ Refractive Abbe Refraction/ Surface curvature gap index (nd) number (vd) reflection 0 Stop Infinity 15.000 — — — 1 First lens 24.533 5.875 1.544 55.9 Refraction 2 126.787 4.805 — — Refraction 3 First absorptive polarizer Infinity 0.100 1.533 56.0 Refraction 4 Reflective polarizer Infinity 0.100 1.533 56.0 Refraction 5 First phase retarder Infinity 0.100 1.533 56.0 Refraction 6 Second lens Infinity 4.413 1.544 55.9 Refraction 7 Third lens -36.214 1.891 1.645 23.400 Refraction 8 Partial-reflective-partial- -72.358 -1.891 — — Reflection transmissive element 9 Second lens -36.214 -4.413 1.544 55.900 Refraction 10 First phase retarder Infinity -0.100 1.533 56.0 Refraction 11 Reflective polarizer Infinity -0.100 1.533 56.0 Refraction 12 Reflective polarizer Infinity 0.100 — — Reflection 13 First phase retarder Infinity 0.100 1.533 56.0 Refraction 14 Second lens Infinity 4.413 1.544 55.900 Refraction 15 Third lens -36.214 1.891 1.544 55.9 Refraction 16 Partial-reflective-partial- -72.358 6.516 — — Refraction transmissive element 17 Second phase retarder Infinity 0.100 1.533 56.0 Refraction 18 Second absorptive Infinity 0.100 1.533 56.0 Refraction polarizer 19 Image source plane Infinity — — — — The reference wavelength is 550 nm.

(57) TABLE-US-00010 TABLE 10 Embodiment 3 Aspheric Coefficients Surface 1 2 6, 10, 14 7, 9, 15 7, 9, 15 8, 16 K: -1.1894E+01 9.0000E+01 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: 6.6598E-05 -5.0930E-05 0.0000E+00 0.0000E+00 0.0000E+00 5.2297E-06 A6: -7.5449E-07 -1.6412E-07 0.0000E+00 0.0000E+00 0.0000E+00 1.7442E-08 A8: 4.3204E-09 -5.4893E-10 0.0000E+00 0.0000E+00 0.0000E+00 -8.2624E-12 A10: -2.0618E-11 -2.6110E-12 0.0000E+00 0.0000E+00 0.0000E+00 2.7055E-13 A12: -1.0590E-13 -8.4544E-15 0.0000E+00 0.0000E+00 0.0000E+00 -1.0573E-15 A14: 1.0188E-15 7.4121E-18 0.0000E+00 0.0000E+00 0.0000E+00 9.4549E-19 A16: 2.0165E-19 7.1377E-19 0.0000E+00 0.0000E+00 0.0000E+00 -1.2703E-20 A18: 7.1422E-22 6.5305E-23 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: -2.0311E-22 -5.0571E-23 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00

(58) TABLE-US-00011 TABLE 11 Embodiment 3 f_1 [mm] 54.48 CA1 [mm] 10.96 TDP1 [mm] 1.73 f_2 [mm] 66.17 CA2 [mm] 11.86 TDP2 [mm] 1.16 f_3 [mm] -114.11 — — — —

(59) TABLE-US-00012 TABLE 12 Embodiment 3 f_1/f 2.75 f_3/f -5.77 R1/R2 0.19 R6/R5 2.00 R6/R2 -0.57 R6/CT3 -38.27 CT3/CT2 0.43 (R1*R2)/ f_1 -27.26 (CT1 + CT2)/CT3 5.44 CA1/TDP1 6.33 R1/CT1 4.18 CA2/TDP2 10.21 f_3/R_1 -4.65 (CA2/TL)*R1 12.13 f_1/f_3 -0.48

Fourth Embodiment

(60) Referring to FIG. 4, an optical lens assembly in accordance with a fourth embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis **490**: a stop **400**, a first lens **410**, a first optical element group **440**, a second lens **420**, a third lens **430**, a partial-reflective-partial-transmissive element **450**, a second optical element group **460** and an image source plane **481**. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

(61) The stop **400** may be located in a position where the user's eyes view an image.

(62) The first lens **410** with positive refractive power includes a visual-side surface **411** and an image source-side surface **412**, the visual-side surface **411** of the first lens **410** is convex in a paraxial region thereof, the image source-side surface **412** of the first lens **410** is concave in a paraxial region thereof, the visual-side surface **411** and the image source-side surface **412** of the first lens **410** are aspheric, and the first lens **410** is made of plastic.

(63) The second lens **420** with positive refractive power includes a visual-side surface **421** and an image source-side surface **422**, the visual-side surface **421** of the second lens **420** is flat in a paraxial region thereof, the image source-side surface **422** of the second lens **420** is convex in a paraxial region thereof, the image source-side surface **422** of the second lens **420** is aspheric, and the second lens **420** is made of plastic.

(64) The third lens **430** with negative refractive power includes a visual-side surface **431** and an image source-side surface **432**, the visual-side surface **431** of the third lens **430** is concave in a paraxial region thereof, the image source-side surface **432** of the third lens **430** is convex in a paraxial region thereof, the visual-side surface **431** and the image source-side surface **432** of the third lens **430** are aspheric, and the third lens **430** is made of plastic.

(65) The first optical element group **440** is disposed on the visual-side surface **421** of the second lens **420** and includes, in order from the visual side to the image source side along the optical axis **490**: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer **141**, the reflective polarizer **142** and the first phase retarder **143** of the first embodiment and will not be explained again.

(66) The partial-reflective-partial-transmissive element **450** is disposed on the image source-side surface **432** of the third lens **430** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **450** for different wavelengths.

(67) The second optical element group **460** is disposed on the image source plane **481** and includes, in order from the visual side to the image source side along the optical axis **490**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

(68) The optical lens assembly works in cooperation with an image source **483** disposed on the image source plane **481**. In the present embodiment, the type of the image source **483** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

(69) Please refer to Tables 13-16, Table 13 shows the detailed optical data of the elements of the optical lens assembly of the fourth embodiment, Table 14 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the fourth embodiment, Table 15 shows the remaining parameters of the optical lens assembly of the fourth embodiment and the values thereof, and the values of the parameters in Tables 13 and 15 meet the conditional formulas of Table 16. In the fourth embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first

embodiment. The definitions of the surfaces in Table 13 can be referred to the relevant description of Table 1 and will not be explained again.

(70) TABLE-US-00013 TABLE 13 Embodiment 4 $f = 19.56$ mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 52.00° Radius of Thickness/ Refractive Abbe Refraction/ Surface curvature gap index (nd) number (vd) reflection 0 Stop Infinity 15.000 — — — 1 First lens 21.291 6.306 1.544 55.9 Refraction 2 43.174 1.025 — — Refraction 3 First absorptive polarizer Infinity 0.100 1.533 56.0 Refraction 4 Reflective polarizer Infinity 0.100 1.533 56.0 Refraction 5 First phase retarder Infinity 0.100 1.533 56.0 Refraction 6 Second lens Infinity 6.269 1.544 55.9 Refraction 7 -21.760 0.428 — — Refraction 8 third lens -27.707 1.740 1.544 55.900 Refraction 9 Partial-reflective-partial- -84.726 -1.740 — — Reflection transmissive element 10 -21.760 -0.428 — — Refraction 11 Second lens -27.707 -6.269 1.544 55.900 Refraction 12 First phase retarder Infinity -0.100 1.533 56.0 Refraction 13 Reflective polarizer Infinity -0.100 1.533 56.0 Refraction 14 Reflective polarizer Infinity 0.100 — — Reflection 15 First phase retarder Infinity 0.100 1.533 56.0 Refraction 16 Second lens Infinity 6.269 1.544 55.900 Refraction 17 -21.760 0.428 — — Refraction 18 Third lens -27.707 1.740 1.544 55.9 Refraction 19 Partial-reflective-partial- -84.726 4.294 — — Refraction transmissive element 20 Second phase retarder Infinity 0.100 1.533 56.0 Refraction 21 Second absorptive Infinity 0.100 1.533 56.0 Refraction polarizer 22 Image source plane Infinity — — — — The reference wavelength is 550 nm.

(71) TABLE-US-00014 TABLE 14 Embodiment 4 Aspheric Coefficients Surface 1 2 6, 12, 16 7, 10, 17 8, 11, 18 9, 19 K: -2.5932E-01 -2.0810E+00 0.0000E+00 -8.1323E-01 -4.6842E-01 8.2000E+00 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: -1.4555E-05 -1.7511E-05 0.0000E+00 2.9577E-06 -5.0664E-05 -2.8280E-06 A6: 3.8026E-08 -1.1557E-07 0.0000E+00 -5.0314E-08 3.9196E-10 3.2122E-08 A8: -8.3422E-10 -6.3251E-10 0.0000E+00 -2.6171E-10 6.2425E-10 1.5667E-10 A10: -4.3414E-12 -2.6505E-12 0.0000E+00 7.8191E-13 3.2496E-13 3.0556E-13 A12: 1.1171E-14 -1.9553E-14 0.0000E+00 1.3185E-14 5.7519E-15 -4.0344E-16 A14: 1.0867E-16 -7.9811E-17 0.0000E+00 6.2014E-17 9.9697E-17 -7.0206E-18 A16: -2.0298E-18 7.0121E-19 0.0000E+00 2.7812E-20 2.6410E-19 9.1281E-20 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00

(72) TABLE-US-00015 TABLE 15 Embodiment 4 f_1 [mm] 69.65 CA1 [mm] 11.32 TDP1 [mm] 2.62 f_2 [mm] 39.76 CA2 [mm] 12.17 TDP2 [mm] 0.26 f_3 [mm] -76.05 — — — —

(73) TABLE-US-00016 TABLE 16 Embodiment 4 f_1/f 3.56 f_3/f -3.89 R1/R2 0.49 R6/R5 3.06 R6/R2 -1.96 R6/CT3 -48.70 CT3/CT2 0.28 (R1*R2)/ f_1 -12.09 (CT1 + CT2)/CT3 7.23 CA1/TDP1 4.33 R1/CT1 3.38 CA2/TDP2 46.80 f_3/R_1 -3.57 (CA2/TL)*R1 12.60 f_1/f_3 -0.92 (R2/CT1)(f_1/f) 24.38 f_2/f 2.03 — —

Fifth Embodiment

(74) Referring to FIG. 5, an optical lens assembly in accordance with a fifth embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis 590: a stop 500, a first lens 510, a second lens 520, a first optical element group 540, a third lens 530, a partial-reflective-partial-transmissive element 550, a second optical element group 560 and an image source plane 581. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

(75) The stop 500 may be located in a position where the user's eyes view an image.

(76) The first lens 510 with positive refractive power includes a visual-side surface 511 and an image source-side surface 512, the visual-side surface 511 of the first lens 510 is convex in a paraxial region thereof, the image source-side surface 512 of the first lens 510 is concave in a paraxial region thereof, the visual-side surface 511 and the image source-side surface 512 of the first lens 510 are aspheric, and the first lens 510 is made of plastic.

(77) The second lens 520 with positive refractive power includes a visual-side surface 521 and an

image source-side surface **522**, the visual-side surface **521** of the second lens **520** is convex in a paraxial region thereof, the image source-side surface **522** of the second lens **520** is concave in a paraxial region thereof, the visual-side surface **521** and the image source-side surface **522** of the second lens **520** are aspheric, and the second lens **520** is made of plastic.

(78) The third lens **530** with positive refractive power includes a visual-side surface **531** and an image source-side surface **532**, the visual-side surface **531** of the third lens **530** is flat in a paraxial region thereof, the image source-side surface **532** of the third lens **530** is convex in a paraxial region thereof, the image source-side surface **532** of the third lens **530** is aspheric, and the third lens **530** is made of plastic.

(79) The first optical element group **540** is disposed on the visual-side surface **531** of the third lens **530** and includes, in order from the visual side to the image source side along the optical axis **590**: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer **141**, the reflective polarizer **142** and the first phase retarder **143** of the first embodiment and will not be explained again.

(80) The partial-reflective-partial-transmissive element **550** is disposed on the image source-side surface **532** of the third lens **530** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **550** for different wavelengths.

(81) The second optical element group **560** is disposed on the image source plane **581** and includes, in order from the visual side to the image source side along the optical axis **590**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

(82) The optical lens assembly works in cooperation with an image source **583** disposed on the image source plane **581**. In the present embodiment, the type of the image source **583** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

(83) Please refer to Tables 17-20, Table 17 shows the detailed optical data of the elements of the optical lens assembly of the fifth embodiment, Table 18 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the fifth embodiment, Table 19 shows the remaining parameters of the optical lens assembly of the fifth embodiment and the values thereof, and the values of the parameters in Tables 17 and 19 meet the conditional formulas of Table 20. In the fifth embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 17 can be referred to the relevant description of Table 1 and will not be explained again.

(84) TABLE-US-00017 TABLE 17 Embodiment 5 $f = 18.17$ mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 56.00° Radius of Thickness/ Refractive Abbe Refraction/ Surface curvature gap index (nd) number (vd) reflection 0 Stop Infinity 15.000 — — — 1 First lens 16.833 4.979 1.544 55.9 Refraction 2 17.373 1.204 — — Refraction 3 Second lens 24.044 4.664 1.544 55.9 Refraction 4 102.772 2.000 — — Refraction 5 First absorptive polarizer Infinity 0.100 1.533 56.0 Refraction 6 Reflective polarizer Infinity 0.100 1.533 56.0 Refraction 7 First phase retarder Infinity 0.100 1.533 56.0 Refraction 8 Third lens Infinity 5.784 1.544 55.900 Refraction 9 Partial-reflective-partial- -66.430 -5.784 — — Reflection transmissive element 10 First phase retarder Infinity -0.100 1.533 56.0 Refraction 11 Reflective polarizer Infinity -0.100 1.533 56.0 Refraction 12 Reflective polarizer Infinity 0.100 — — Reflection 13 First phase retarder Infinity 0.100 1.533 56.0 Refraction 14 Third lens Infinity 5.784 1.544 55.9 Refraction 15 Partial-reflective-partial- -66.430 4.870 — — Refraction transmissive element 16 Second phase retarder Infinity 0.100 1.533 56.0 Refraction 17 Second absorptive Infinity 0.100 1.533 56.0

Refraction polarizer 18 Image source plane Infinity — — — — The reference wavelength is 550 nm.

(85) TABLE-US-00018 TABLE 18 Embodiment 5 Aspheric Coefficients Surface 1 2 3 4 8, 10, 14 9, 15 K: $-2.4483\text{E}-01$ $3.6440\text{E}-02$ $-5.3088\text{E}-02$ $-6.3630\text{E}+01$ $0.0000\text{E}+00$ $-1.2456\text{E}+00$
A2: $0.0000\text{E}+00$ $0.0000\text{E}+00$ $0.0000\text{E}+00$ $0.0000\text{E}+00$ $0.0000\text{E}+00$ $0.0000\text{E}+00$ A4:
 $-1.3939\text{E}-05$ $-1.3675\text{E}-05$ $-3.8894\text{E}-06$ $-8.0253\text{E}-06$ $0.0000\text{E}+00$ $5.9760\text{E}-06$ A6:
 $2.3663\text{E}-09$ $-7.1810\text{E}-08$ $-8.5553\text{E}-09$ $-2.6639\text{E}-08$ $0.0000\text{E}+00$ $1.2082\text{E}-08$ A8:
 $-3.1998\text{E}-10$ $-3.7038\text{E}-10$ $-2.5552\text{E}-11$ $4.1260\text{E}-11$ $0.0000\text{E}+00$ $-1.1398\text{E}-11$ A10:
 $-1.0722\text{E}-12$ $-1.2803\text{E}-12$ $4.4043\text{E}-13$ $4.1232\text{E}-13$ $0.0000\text{E}+00$ $8.5732\text{E}-13$ A12:
 $5.0972\text{E}-15$ $-2.4952\text{E}-15$ $-1.4351\text{E}-16$ $9.0535\text{E}-16$ $0.0000\text{E}+00$ $-4.6428\text{E}-16$ A14:
 $4.5081\text{E}-18$ $-8.4687\text{E}-18$ $-6.0146\text{E}-18$ $4.0754\text{E}-19$ $0.0000\text{E}+00$ $-4.7947\text{E}-17$ A16:
 $-8.9487\text{E}-20$ $1.2044\text{E}-20$ $6.2660\text{E}-20$ $-8.9767\text{E}-23$ $0.0000\text{E}+00$ $2.7543\text{E}-19$ A18:
 $0.0000\text{E}+00$ $0.0000\text{E}+00$ $0.0000\text{E}+00$ $0.0000\text{E}+00$ $0.0000\text{E}+00$ $0.0000\text{E}+00$ A20: $0.0000\text{E}+00$
 $0.0000\text{E}+00$ $0.0000\text{E}+00$ $0.0000\text{E}+00$ $0.0000\text{E}+00$ $0.0000\text{E}+00$

(86) TABLE-US-00019 TABLE 19 Embodiment 5 f1 [mm] 232.24 CA1 [mm] 13.61 TDP1 [mm] 5.47 f2 [mm] 56.18 CA2 [mm] 13.66 TDP2 [mm] 4.88 f3 [mm] 121.38 — — — —

(87) TABLE-US-00020 TABLE 20 Embodiment 5 f1/f 12.78 f3/f 6.68 R1/R2 0.97 R6/R5 0.00 R6/R2 -3.82 R6/CT3 -11.49 CT3/CT2 1.24 (R1*R2)/f1 2.41 (CT1 + CT2)/CT3 1.67 CA1/TDP1 2.49 R1/CT1 3.38 CA2/TDP2 2.80 f3/R1 7.21 (CA2/TL)*R1 9.58 f1/f3 1.91 (R2/CT1)(f1/f) 44.59 f2/f 3.09 — —

Sixth Embodiment

(88) Referring to FIG. 6, an optical lens assembly in accordance with a sixth embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis 690: a stop 600, a first lens 610, a second lens 620, a first optical element group 640, a third lens 630, a partial-reflective-partial-transmissive element 650, a second optical element group 660 and an image source plane 681. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

(89) The stop 600 may be located in a position where the user's eyes view an image.

(90) The first lens 610 with positive refractive power includes a visual-side surface 611 and an image source-side surface 612, the visual-side surface 611 of the first lens 610 is convex in a paraxial region thereof, the image source-side surface 612 of the first lens 610 is convex in a paraxial region thereof, the visual-side surface 611 and the image source-side surface 612 of the first lens 610 are aspheric, and the first lens 610 is made of plastic.

(91) The second lens 620 with negative refractive power includes a visual-side surface 621 and an image source-side surface 622, the visual-side surface 621 of the second lens 620 is concave in a paraxial region thereof, the image source-side surface 622 of the second lens 620 is convex in a paraxial region thereof, the visual-side surface 621 and the image source-side surface 622 of the second lens 620 are aspheric, and the second lens 620 is made of plastic.

(92) The third lens 630 with positive refractive power includes a visual-side surface 631 and an image source-side surface 632, the visual-side surface 631 of the third lens 630 is flat in a paraxial region thereof, the image source-side surface 632 of the third lens 630 is convex in a paraxial region thereof, the image source-side surface 632 of the third lens 630 is spherical, and the third lens 630 is made of plastic.

(93) The first optical element group 640 is disposed on the visual-side surface 631 of the third lens 630 and includes, in order from the visual side to the image source side along the optical axis 690: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer 141, the reflective polarizer 142 and the first phase retarder 143 of the first embodiment and will not be explained again.

(94) The partial-reflective-partial-transmissive element 650 is disposed on the image source-side

surface **632** of the third lens **630** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **650** for different wavelengths.

(95) The second optical element group **660** is disposed on the image source plane **681** and includes, in order from the visual side to the image source side along the optical axis **690**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

(96) The optical lens assembly works in cooperation with an image source **683** disposed on the image source plane **681**. In the present embodiment, the type of the image source **683** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

(97) Please refer to Tables 21-24, Table 21 shows the detailed optical data of the elements of the optical lens assembly of the sixth embodiment, Table 22 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the sixth embodiment, Table 23 shows the remaining parameters of the optical lens assembly of the sixth embodiment and the values thereof, and the values of the parameters in Tables 21 and 23 meet the conditional formulas of Table 24. In the sixth embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 21 can be referred to the relevant description of Table 1 and will not be explained again.

(98) TABLE-US-00021 TABLE 21 Embodiment 6 $f = 18.14$ mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 60.52° Radius of Thickness/ Refractive Abbe Refraction/ Surface curvature gap Index (nd) number (vd) reflection 0 Stop Infinity 15.000 — — — 1 First lens 67.587 8.308 1.544 55.9 Refraction 2 -22.871 0.964 — — Refraction 3 Second lens -13.410 2.710 1.544 55.9 Refraction 4 -19.512 0.100 — — Refraction 5 First absorptive polarizer Infinity 0.100 1.533 56.0 Refraction 6 Reflective polarizer Infinity 0.100 1.533 56.0 Refraction 7 First phase retarder Infinity 0.100 1.533 56.0 Refraction 8 Third lens Infinity 10.418 1.544 55.900 Refraction 9 Partial-reflective-partial- -69.351 -10.418 — — Reflection transmissive element 10 First phase retarder Infinity -0.100 1.533 56.0 Refraction 11 Reflective polarizer Infinity -0.100 1.533 56.0 Refraction 12 Reflective polarizer Infinity 0.100 — — Reflection 13 First phase retarder Infinity 0.100 1.533 56.0 Refraction 14 Third lens Infinity 10.418 1.544 55.9 Refraction 15 Partial-reflective-partial- -69.351 1.000 — — Refraction transmissive element 16 Second phase retarder Infinity 0.100 1.533 56.0 Refraction 17 Second absorptive Infinity 0.100 1.533 56.0 Refraction polarizer 18 Image source plane Infinity — — — — The reference wavelength is 550 nm.

(99) TABLE-US-00022 TABLE 22 Embodiment 6 Aspheric Coefficients Surface 1 2 3 4 8, 10, 14 9, 15 K: 0.0000E+00 0.0000E+00 -2.0627E-01 0.0000E+00 0.0000E+00 0.0000E+00 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: -1.1715E-05 -4.0874E-05 5.8026E-05 6.4153E-05 0.0000E+00 0.0000E+00 A6: 3.7663E-11 -8.0525E-08 1.6400E-07 1.0691E-07 0.0000E+00 0.0000E+00 A8: -7.2412E-10 7.0159E-11 2.0777E-10 -1.9260E-10 0.0000E+00 0.0000E+00 A10: -7.5971E-12 2.4966E-12 6.1292E-12 -8.0932E-13 0.0000E+00 0.0000E+00 A12: -1.4095E-14 1.8405E-14 2.5661E-15 7.6759E-16 0.0000E+00 0.0000E+00 A14: 4.4519E-16 -2.5695E-16 -1.3272E-16 1.2401E-17 0.0000E+00 0.0000E+00 A16: -3.9216E-18 0.0000E+00 5.8205E-19 -1.4243E-20 0.0000E+00 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00

(100) TABLE-US-00023 TABLE 23 Embodiment 6 f_1 [mm] 32.27 CA1 [mm] 11.54 TDP1 [mm] 0.10 f_2 [mm] -80.09 CA2 [mm] 12.96 TDP2 [mm] 5.73 f_3 [mm] 126.72 — — — —

(101) TABLE-US-00024 TABLE 24 Embodiment 6 f1/f 1.78 f3/f 6.99 R1/R2 -2.96 R6/R5 0.00 R6/R2 3.03 R6/CT3 -6.66 CT3/CT2 3.84 (R1*R2)/f1 -12.20 (CT1 + CT2)/CT3 1.06 CA1/TDP1 114.31 R1/CT1 8.14 CA2/TDP2 2.26 f3/R1 1.87 (CA2/TL)*R1 36.50 f1/f3 0.25 (R2/CT1)(f1/f) -4.90 f2/f -4.41 — —

Seventh Embodiment

(102) Referring to FIG. 7, an optical lens assembly in accordance with a seventh embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis **790**: a stop **700**, a first lens **710**, a second lens **720**, a first optical element group **740**, a third lens **730**, a partial-reflective-partial-transmissive element **750**, a second optical element group **760** and an image source plane **781**. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

(103) The stop **700** may be located in a position where the user's eyes view an image.

(104) The first lens **710** with positive refractive power includes a visual-side surface **711** and an image source-side surface **712**, the visual-side surface **711** of the first lens **710** is concave in a paraxial region thereof, the image source-side surface **712** of the first lens **710** is convex in a paraxial region thereof, the visual-side surface **711** and the image source-side surface **712** of the first lens **710** are aspheric, and the first lens **710** is made of plastic.

(105) The second lens **720** with negative refractive power includes a visual-side surface **721** and an image source-side surface **722**, the visual-side surface **721** of the second lens **720** is concave in a paraxial region thereof, the image source-side surface **722** of the second lens **720** is convex in a paraxial region thereof, the visual-side surface **721** and the image source-side surface **722** of the second lens **720** are aspheric, and the second lens **720** is made of plastic. The second lens **720** and the first lens **710** together form a cemented doublet lens.

(106) The third lens **730** with positive refractive power includes a visual-side surface **731** and an image source-side surface **732**, the visual-side surface **731** of the third lens **730** is flat in a paraxial region thereof, the image source-side surface **732** of the third lens **730** is convex in a paraxial region thereof, the image source-side surface **732** of the third lens **730** is spherical, and the third lens **730** is made of plastic.

(107) The first optical element group **740** is disposed on the visual-side surface **731** of the third lens **730** and includes, in order from the visual side to the image source side along the optical axis **790**: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer **141**, the reflective polarizer **142** and the first phase retarder **143** of the first embodiment and will not be explained again.

(108) The partial-reflective-partial-transmissive element **750** is disposed on the image source-side surface **732** of the third lens **730** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **750** for different wavelengths.

(109) The second optical element group **760** is disposed on the image source plane **781** and includes, in order from the visual side to the image source side along the optical axis **790**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

(110) The optical lens assembly works in cooperation with an image source **783** disposed on the image source plane **781**. In the present embodiment, the type of the image source **783** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

(111) Please refer to Tables 25-28, Table 25 shows the detailed optical data of the elements of the optical lens assembly of the seventh embodiment, Table 26 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the seventh embodiment, Table 27 shows the

remaining parameters of the optical lens assembly of the seventh embodiment and the values thereof, and the values of the parameters in Tables 25 and 27 meet the conditional formulas of Table 28. In the seventh embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 25 can be referred to the relevant description of Table 1 and will not be explained again.

(112) TABLE-US-00025 TABLE 25 Embodiment 7 $f = 19.57$ mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 56.00° Radius of Thickness/ Refractive Abbe Refraction/ Surface curvature gap index (nd) number (vd) reflection 0 Stop Infinity 15.000 — — — 1 First lens -87.167 7.762 1.544 55.9 Refraction 2 -16.051 0.000 — — Refraction 3 Second lens -16.051 2.000 1.645 23.4 Refraction 4 -23.678 0.100 — — Refraction 5 First absorptive polarizer Infinity 0.100 1.533 56.0 Refraction 6 Reflective polarizer Infinity 0.100 1.533 56.0 Refraction 7 First phase retarder Infinity 0.100 1.533 56.0 Refraction 8 Third lens Infinity 12.638 1.544 55.900 Refraction 9 Partial-reflective-partial- -81.316 -12.638 — — Reflection transmissive element 10 First phase retarder Infinity -0.100 1.533 56.0 Refraction 11 Reflective polarizer Infinity -0.100 1.533 56.0 Refraction 12 Reflective polarizer Infinity 0.100 — — Reflection 13 First phase retarder Infinity 0.100 1.533 56.0 Refraction 14 Third lens Infinity 12.638 1.544 55.9 Refraction 15 Partial-reflective-partial- -81.316 1.000 — — Refraction transmissive element 16 Second phase retarder Infinity 0.100 1.533 56.0 Refraction 17 Second absorptive Infinity 0.100 1.533 56.0 Refraction polarizer 18 Image source plane Infinity — — — — The reference wavelength is 550 nm.

(113) TABLE-US-00026 TABLE 26 Embodiment 7 Aspheric Coefficients Surface 1 2 3 4 8, 10, 14 9, 15 K: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: -2.3306E-05 -1.9255E-04 -1.9255E-04 -3.5158E-05 0.0000E+00 0.0000E+00 A6: -2.6725E-07 1.4129E-06 1.4129E-06 1.2565E-07 0.0000E+00 0.0000E+00 A8: 1.9968E-09 -3.9152E-09 -3.9152E-09 -4.6121E-10 0.0000E+00 0.0000E+00 A10: -2.3610E-11 -2.9871E-11 -2.9871E-11 -4.1219E-12 0.0000E+00 0.0000E+00 A12: -4.9339E-14 3.8751E-14 3.8751E-14 8.8672E-15 0.0000E+00 0.0000E+00 A14: 1.9228E-15 8.2701E-16 8.2701E-16 1.0410E-16 0.0000E+00 0.0000E+00 A16: -1.1411E-17 0.0000E+00 0.0000E+00 -4.7844E-19 0.0000E+00 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00

(114) TABLE-US-00027 TABLE 27 Embodiment 7 f_1 [mm] 34.61 CA1 [mm] 10.11 TDP1 [mm] 1.13 f_2 [mm] -85.66 CA2 [mm] 11.37 TDP2 [mm] 6.37 f_3 [mm] 148.59 — — — —

(115) TABLE-US-00028 TABLE 28 Embodiment 7 f_1/f 1.77 f_3/f 7.59 R1/R2 5.43 R6/R5 0.00 R6/R2 5.07 R6/CT3 -6.43 CT3/CT2 6.32 $(R1 \cdot R2)/f_1$ 9.42 $(CT1 + CT2)/CT3$ 0.77 CA1/TDP1 8.94 R1/CT1 -11.23 CA2/TDP2 1.78 $f_3/R1$ -1.70 $(CA2/TL) \cdot R1$ -41.28 f_1/f_3 0.23 $(R2/CT1)(f_1/f)$ -3.66 f_2/f -4.38 — —

Eighth Embodiment

(116) Referring to FIG. 8, an optical lens assembly in accordance with an eighth embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis **890**: a stop **800**, a first lens **810**, a second lens **820**, a first optical element group **840**, a third lens **830**, a partial-reflective-partial-transmissive element **850**, a second optical element group **860** and an image source plane **881**. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

(117) The stop **800** may be located in a position where the user's eyes view an image.

(118) The first lens **810** with positive refractive power includes a visual-side surface **811** and an image source-side surface **812**, the visual-side surface **811** of the first lens **810** is convex in a paraxial region thereof, the image source-side surface **812** of the first lens **810** is concave in a

paraxial region thereof, the visual-side surface **811** and the image source-side surface **812** of the first lens **810** are aspheric, and the first lens **810** is made of plastic.

(119) The second lens **820** with positive refractive power includes a visual-side surface **821** and an image source-side surface **822**, the visual-side surface **821** of the second lens **820** is convex in a paraxial region thereof, the image source-side surface **822** of the second lens **820** is flat in a paraxial region thereof, the visual-side surface **821** of the second lens **820** is aspheric, and the second lens **820** is made of plastic.

(120) The third lens **830** with positive refractive power includes a visual-side surface **831** and an image source-side surface **832**, the visual-side surface **831** of the third lens **830** is convex in a paraxial region thereof, the image source-side surface **832** of the third lens **830** is convex in a paraxial region thereof, the visual-side surface **831** and the image source-side surface **832** of the third lens **830** are aspheric, and the third lens **830** is made of plastic.

(121) The first optical element group **840** is disposed on the image source-side surface **822** of the second lens **820** and includes, in order from the visual side to the image source side along the optical axis **890**: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer **141**, the reflective polarizer **142** and the first phase retarder **143** of the first embodiment and will not be explained again.

(122) The partial-reflective-partial-transmissive element **850** is disposed on the image source-side surface **832** of the third lens **830** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **850** for different wavelengths.

(123) The second optical element group **860** is disposed on the image source plane **881** and includes, in order from the visual side to the image source side along the optical axis **890**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

(124) The optical lens assembly works in cooperation with an image source **883** disposed on the image source plane **881**. In the present embodiment, the type of the image source **883** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

(125) Please refer to Tables 29-32, Table 29 shows the detailed optical data of the elements of the optical lens assembly of the eighth embodiment, Table 30 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the eighth embodiment, Table 31 shows the remaining parameters of the optical lens assembly of the eighth embodiment and the values thereof, and the values of the parameters in Tables 29 and 31 meet the conditional formulas of Table 32. In the eighth embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 29 can be referred to the relevant description of Table 1 and will not be explained again.

(126) TABLE-US-00029 TABLE 29 Embodiment 8 $f = 18.19$ mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 57.71° Radius of Thickness/ Refractive Abbe Refraction/ Surface curvature gap index (nd) number (vd) reflection 0 Stop Infinity 15.000 — — — 1 First lens 34.173 6.274 1.544 55.9 Refraction 2 42.937 4.857 — — Refraction 3 Second lens 110.714 2.135 1.544 55.9 Refraction 4 First absorptive polarizer Infinity 0.100 1.533 56.0 Refraction 5 Reflective polarizer Infinity 0.100 1.533 56.0 Refraction 6 First phase retarder Infinity 0.100 1.533 56.0 Refraction 7 Infinity 1.890 — — Refraction 8 Third lens 129.300 7.063 1.544 55.900 Refraction 9 Partial-reflective-partial- -68.753 -7.063 — — Reflection transmissive element 10 129.300 -1.890 — — Refraction 11 First phase retarder Infinity -0.100 1.533 56.0 Refraction 12 Reflective polarizer Infinity -0.100 1.533 56.0 Refraction 13 Reflective polarizer Infinity 0.100 —

— Reflection 14 First phase retarder Infinity 0.100 1.533 56.0 Refraction 15 Infinity 1.890 — —
Refraction 16 Third lens 129.300 7.063 1.544 55.9 Refraction 17 Partial-reflective-partial- -68.753
2.280 — — Refraction transmissive element 18 Second phase retarder Infinity 0.100 1.533 56.0
Refraction 19 Second absorptive Infinity 0.100 1.533 56.0 Refraction polarizer 20 Image source
plane Infinity — — — The reference wavelength is 550 nm.

(127) TABLE-US-00030 TABLE 30 Embodiment 8 Aspheric Coefficients Surface 1 2 3 4 8, 10, 16
9, 17 K: -5.3525E+00 -7.5221E+00 -1.7631E+01 0.0000E+00 -3.0407E+01 -1.1050E+00 A2:
0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: -6.9119E-06
-7.6849E-06 -7.9976E-08 0.0000E+00 -1.4962E-06 3.5987E-07 A6: -4.9302E-08
-2.7343E-08 1.1354E-08 0.0000E+00 -8.2575E-10 1.0984E-09 A8: -8.7928E-11 -1.0237E-10
9.2473E-12 0.0000E+00 -7.7597E-14 2.3102E-12 A10: -6.0233E-13 -2.3540E-13
-1.2075E-14 0.0000E+00 4.5237E-15 1.7381E-15 A12: -1.6016E-15 -2.0726E-17
-5.7442E-17 0.0000E+00 -1.8839E-18 2.8049E-18 A14: 1.8562E-17 1.6676E-18
-1.3546E-19 0.0000E+00 -4.6076E-20 1.3546E-20 A16: -8.8664E-20 -4.9534E-21
3.1761E-22 0.0000E+00 3.2783E-22 2.7089E-23 A18: 0.0000E+00 0.0000E+00 0.0000E+00
0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00
0.0000E+00 0.0000E+00

(128) TABLE-US-00031 TABLE 31 Embodiment 8 f1 [mm] 244.11 CA1 [mm] 14.03 TDP1 [mm]
1.45 f2 [mm] 202.30 CA2 [mm] 16.37 TDP2 [mm] 0.70 f3 [mm] 83.06 — — — —

(129) TABLE-US-00032 TABLE 32 Embodiment 8 f1/f 13.42 f3/f 4.57 R1/R2 0.80 R6/R5 0.00
R6/R2 -1.60 R6/CT3 -9.73 CT3/CT2 3.31 (R1*R2)/f1 17.66 (CT1 + CT2)/CT3 1.19 CA1/TDP1
9.67 R1/CT1 5.45 CA2/TDP2 23.46 f3/R1 2.43 (CA2/TL)*R1 22.38 f1/f3 2.94 (R2/CT1)(f1/f)
91.85 f2/f 11.12 — —

(130) For the optical lens assembly in the present invention, the lenses can be made of plastic or glass. If the lens is made of plastic, it is conducive to reducing the manufacturing cost. If the lens is made of glass, it is conducive to enhancing the degree of freedom in the arrangement of refractive power of the optical lens assembly.

(131) For the optical lens assembly in the present invention, the aspheric surface can have any profile shape other than the profile shape of a spherical surface, so more variables can be used in the design of aspheric surfaces (than spherical surfaces), which is conducive to reducing the aberration and the number of lenses, as well as the total length of the optical lens assembly.

(132) For the optical lens assembly in the present invention, if the surface shape of a respective lens surface of a respective lens with refractive power is convex and the location of the convex portion of the respective lens surface of the respective lens is not defined, the convex portion is typically located in a paraxial region of the respective lens surface of the respective lens. If the surface shape of a respective lens surface of a respective lens is concave and the location of the concave portion of the respective lens surface of the respective lens is not defined, the concave portion is typically located in a paraxial region of the respective lens surface of the respective lens.

(133) For the optical lens assembly in the present invention, the maximum effective radius of the lens surface is usually a radius of the effective optical region of the lens surface (i.e., a region which is not subjected to any surface treatment or extinction processing or is not provided with any shade).

(134) The optical lens assembly of the present invention can be used in head-mounted electronic devices as required. FIG. 9 shows a head-mounted electronic device in accordance with an embodiment of the present invention. The head-mounted electronic device 9 is a head-mounted display device using, but not limited to, virtual reality (VR) technology, and includes a housing 910, an optical module 920, an image source 930 and a controller 940.

(135) The optical module 920 corresponds to the left and right eyes of the user. The optical module 920 includes an optical lens assembly described in any one of the first to eighth embodiments.

(136) The image source 930 can be an image source described in any one of the first to eighth

embodiments. The image source **930** corresponds to the left and right eyes of the user, and the type of the image source **930** may be an OLED display, a LED display, a liquid crystal display, or other display, but is not limited thereto.

(137) The controller **940** is electrically connected to the image source **930**, so as to control the image source **930** to display an image, whereby the head-mounted electronic device **9** can project the image to the eyes of the user.

Claims

1. An optical lens assembly, comprising: an optical element group; and in order from a visual side to an image source side: a first lens with positive refractive power, a second lens with refractive power, a third lens with refractive power, and an image source-side surface of the third lens being convex in a paraxial region thereof, and a partial-reflective-partial-transmissive element; wherein the optical element group is disposed between the first lens and the third lens, and includes, in order from the visual side to the image source side: an absorptive polarizer, a reflective polarizer and a phase retarder, a radius of curvature of an image source-side surface of the first lens is R_2 , a thickness of the first lens along an optical axis is CT_1 , a focal length of the first lens is f_1 , a focal length of the optical lens assembly is f , and the following condition is satisfied: $-5.88 < (R_2/CT_1) * (f_1/f) < 110.22$.
2. The optical lens assembly as claimed in claim 1, wherein a focal length of the third lens is f_3 , the focal length of the optical lens assembly is f , and the following condition is satisfied: $-6.92 < f_3/f < 23.80$.
3. The optical lens assembly as claimed in claim 1, wherein a focal length of the third lens is f_3 , a radius of curvature of a visual-side surface of the first lens is R_1 , and the following condition is satisfied: $-5.58 < f_3/R_1 < 27.18$.
4. The optical lens assembly as claimed in claim 1, wherein a maximum effective radius of a visual-side surface of the first lens is CA_1 , an absolute value of a displacement in parallel to the optical axis from an intersection between the visual-side surface of the first lens and the optical axis to the maximum effective radius position on the visual-side surface of the first lens is TDP_1 , and the following condition is satisfied: $1.76 < CA_1/TDP_1 < 137.17$.
5. The optical lens assembly as claimed in claim 1, wherein a maximum effective radius of the image source-side surface of the first lens is CA_2 , an absolute value of a displacement in parallel to the optical axis from an intersection between the image source-side surface of the first lens and the optical axis to the maximum effective radius position on the image source-side surface of the first lens is TDP_2 , and the following condition is satisfied: $1.43 < CA_2/TDP_2 < 56.16$.
6. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of the image source-side surface of the third lens is R_6 , a thickness of the third lens along the optical axis is CT_3 , and the following condition is satisfied: $-58.44 < R_6/CT_3 < -5.15$.
7. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of a visual-side surface of the first lens is R_1 , the radius of curvature of the image source-side surface of the first lens is R_2 , the focal length of the first lens is f_1 , and the following condition is satisfied: $-32.71 \text{ mm} < (R_1 * R_2)/f_1 < 60.84 \text{ mm}$.
8. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of a visual-side surface of the first lens is R_1 , the radius of curvature of the image source-side surface of the first lens is R_2 , and the following condition is satisfied: $-3.55 < R_1/R_2 < 6.52$.
9. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of the image source-side surface of the third lens is R_6 , the radius of curvature of the image source-side surface of the first lens is R_2 , and the following condition is satisfied: $-4.59 < R_6/R_2 < 6.08$.
10. The optical lens assembly as claimed in claim 1, wherein a thickness of the second lens along the optical axis is CT_2 , a thickness of the third lens along the optical axis is CT_3 , and the following

condition is satisfied: $0.22 < CT3/CT2 < 7.58$.

11. The optical lens assembly as claimed in claim 1, wherein the thickness of the first lens along the optical axis is CT1, a thickness of the second lens along the optical axis is CT2, a thickness of the third lens along the optical axis is CT3, and the following condition is satisfied:

$$0.62 < (CT1 + CT2)/CT3 < 8.67.$$

12. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of a visual-side surface of the first lens is R1, the thickness of the first lens along an optical axis is CT1, and the following condition is satisfied: $-13.48 < R1/CT1 < 9.76$.

13. The optical lens assembly as claimed in claim 1, wherein the focal length of the first lens is f1, the focal length of the optical lens assembly is f, and the following condition is satisfied:

$$1.42 < f1/f < 16.1.$$

14. The optical lens assembly as claimed in claim 1, wherein a maximum effective radius of the image source-side surface of the first lens is CA2, a distance from a visual-side surface of the first lens to an image source plane along the optical axis is TL, a radius of curvature of the visual-side surface of the first lens is R1, and the following condition is satisfied: -49.53

$$\text{mm} < (CA2/TL) * R1 < 43.8 \text{ mm}.$$

15. A head-mounted electronic device, comprising: a housing; an optical lens assembly disposed in the housing; an image source disposed on an image source plane of the optical lens assembly in the housing; and a controller disposed in the housing and electrically connected to the image source; wherein the optical lens assembly comprising: an optical element group; and in order from a visual side to an image source side: a first lens with positive refractive power, a second lens with refractive power, a third lens with refractive power, and an image source-side surface of the third lens being convex in a paraxial region thereof, and a partial-reflective-partial-transmissive element; wherein the optical element group is disposed between the first lens and the third lens, and includes, in order from the visual side to the image source side: an absorptive polarizer, a reflective polarizer and a phase retarder, a radius of curvature of an image source-side surface of the first lens is R2, a thickness of the first lens along an optical axis is CT1, a focal length of the first lens is f1, a focal length of the optical lens assembly is f, and the following condition is satisfied:

$$-5.88 < (R2/CT1) * (f1/f) < 110.22.$$

16. The head-mounted electronic device as claimed in claim 15, wherein a focal length of the third lens is f3, the focal length of the optical lens assembly is f, and the following condition is satisfied: $-6.92 < f3/f < 23.80$.

17. The head-mounted electronic device as claimed in claim 15, wherein a maximum effective radius of a visual-side surface of the first lens is CA1, an absolute value of a displacement in parallel to the optical axis from an intersection between the visual-side surface of the first lens and the optical axis to the maximum effective radius position on the visual-side surface of the first lens is TDP1, and the following condition is satisfied: $1.76 < CA1/TDP1 < 137.17$.

18. The head-mounted electronic device as claimed in claim 15, wherein a radius of curvature of a visual-side surface of the first lens is R1, the radius of curvature of the image source-side surface of the first lens is R2, the focal length of the first lens is f1, and the following condition is satisfied:

$$-32.71 \text{ mm} < (R1 * R2)/f1 < 60.84 \text{ mm}.$$

19. The head-mounted electronic device as claimed in claim 15, wherein the thickness of the first lens along the optical axis is CT1, a thickness of the second lens along the optical axis is CT2, a thickness of the third lens along the optical axis is CT3, and the following condition is satisfied:

$$0.62 < (CT1 + CT2)/CT3 < 8.67.$$

20. The head-mounted electronic device as claimed in claim 15, wherein a maximum effective radius of the image source-side surface of the first lens is CA2, a distance from a visual-side surface of the first lens to the image source plane along the optical axis is TL, a radius of curvature of the visual-side surface of the first lens is R1, and the following condition is satisfied: -49.53

$$\text{mm} < (CA2/TL) * R1 < 43.8 \text{ mm}.$$

